VeriQ

Finds the cause, protects what you chose, and proves it.

Intent-aware home broadband QoS with measurable outcomes, bounded adaptation and explainable recovery · Project description

Tata Elxsi TELIPORT Season 4 · Case Study 3 (Media & Communications): Adaptive QoS Engine for Mixed Home Broadband Traffic · Team <Team Name>, <College>

Item	Status (12 Oct 2026)
Design	v4.1 (12 Oct 2026): v3.3 core locked + three refinements (action checker, mark plumbing, evidence-graded diagnosis). Authoritative spec: docs/09 (Appendix A = contracts); change record: docs/10
Measured so far	One preliminary Day-0 sandbox run (Section 8.1; raw files in evidence/). Every other number is a **target**, a published default or someone else's published result, and is labelled as such.
Next milestone	First reproducible call-plus-bulk experiment: FIFO vs SQM-CAKE vs cake-autorate vs VeriQ

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1. Executive summary

The problem. Homes run video calls, games, streaming, downloads and cloud backups on one broadband line. When a bulk transfer fills the oversized buffer in front of the narrowest link, every interactive packet waits behind it. Calls freeze and games lag even on fast plans. Static priority rules cannot fix this: traffic is encrypted, the line's capacity changes, and the queue usually sits in equipment the home router does not control.

Our solution. VeriQ is a controller that runs on the home gateway:

- It shapes just below the line's real capacity, so the queue forms where a modern AQM (CAKE) keeps it short and fair.
- It tracks that capacity as it moves.
- It **finds the cause** before acting: a WAN drop, a remote server, Wi-Fi or a bad measurement.
- It honours time-limited user intents ('protect my work call for 45 minutes').
- It writes an evidence-backed **receipt** for every change: improved, degraded or inconclusive.
- It never reads content, and it starts in a shadow mode that only records what it would do. Classifier abstention alone never grants preference. A valid authenticated lease may grant preference to UNKNOWN traffic, subject to eligibility and operating-state checks.

Why it is different.

- **Diagnosis before action.** Designed so that a slow far-away server does not trigger a wasteful speed cut. Measured as the rate of unnecessary interventions (our primary research question).
- **Real user outcomes.** We count real WebRTC video freezes and DASH stalls, not just ping. IETF Quality-of-Outcome scores are computed offline.
- **Honest ML.** Apps' own NQB/L4S marks come first. The classifier is tested by flow and by time, with shortcut features removed, and abstains when unsure.
- **Honest comparison (planned).** To be tested against tuned SQM-CAKE, cake-autorate and QoSmate with autorate, plus ablations; run-level statistics with confidence intervals. Ties and losses will be reported.
- **Auditable control.** A deterministic action checker admits or rejects every change before it reaches the kernel, and every decision leaves a receipt.

Value. Households get smoother calls and games without buying hardware. Operators get fewer 'my internet is slow' support contacts and standards-aligned QoE telemetry (TR-471/RFC 9946 capacity tests, TR-181/USP management). Tata Elxsi gets a software accelerator for its RDK-B broadband engineering services, consistent with its intent-based, closed-loop NEURON philosophy.

> **Evidence so far (emulated)**
>
> In our Day-0 lab (20/5 Mbps line, 300 ms modem buffer), a cloud-backup-style upload added **266 ms** of 95th-percentile delay to everything else on a standard router. A plain speed limiter cut that to **40 ms**. CAKE and VeriQ arms run next on a full Linux kernel.

2. Problem overview

2.1 What Tata Elxsi asks (Case Study 3)

A residential connection must carry video conferencing, gaming, voice, adaptive video, downloads, software updates and cloud backup **at the same time**. The case makes four observations:

- Static priority rules cannot always tell which activity matters at a particular moment.
- High throughput is insufficient if large queues add delay and jitter.
- The controller must protect interactive applications while bulk traffic keeps making reasonable progress.
- Teams must build a **technically credible reference implementation** with openly documented interfaces and measurable experiments, and state clearly where the prototype ends and a production solution begins.

2.2 The real pain points

Pain point	What actually happens	Why naive fixes fail
Bufferbloat	TCP bulk flows fill oversized modem/ISP buffers; every call/game packet waits behind them	Prioritising at the router does nothing if the router is not the bottleneck
Encrypted traffic	Most traffic is QUIC/TLS on port 443; many apps don't set DSCP	Port rules lose precision and DPI is invasive; headers and timing still carry useful metadata, and fair queuing stays effective
Moving capacity	DOCSIS contention, 5G FWA radio, Wi-Fi interference change the real rate minute to minute	A fixed shaper is either too high (bloat returns) or too low (wasted bandwidth)
Uplink congestion	Home uplinks are slow; backups and uploads cause the worst delay	Download-only or fixed-rate setups miss it; mature tools (SQM, QoSmate) shape both directions
Starvation risk	Strict priority can stop downloads and updates completely	Violates fairness
No verification	QoS is set and forgotten; nobody checks whether calls improved	Ping graphs are not user experience
Wrong diagnosis	Slowness can come from the WAN, a remote server or Wi-Fi	Cutting speed for a remote or Wi-Fi problem wastes capacity and helps no one
Operator cost	'Slow internet' complaints even when speed tests pass	Speed tests measure throughput, not responsiveness

2.3 Illustrative scenarios from the case

1. A large ISO download starts during a video conference: limit the bulk queue, protect call latency.
2. WAN bandwidth drops from 100 to 20 Mbps: recalculate shaping instead of building a large queue.
3. Three TVs stream while a gaming device needs low latency: balance service quality across the household.

2.4 Constraints and design boundaries (from the case)

- Do not decrypt traffic; use Linux traffic control or another open, inspectable mechanism.
- Support IPv4 and, where available, IPv6.

- Provide policy rollback and prevent starvation of low-priority traffic.
- Classifiers must expose confidence and allow correction of misclassification.
- Document third-party licences; include generated data and impairment settings so results reproduce.
- Label emulated hardware capabilities as emulation; never commit real secrets.
- Automated remediation must be bounded, observable, reversible, and fail to a known safe state.
- Where AI is used, compare it with a deterministic baseline and show a measurable improvement, not merely a conversational interface.

2.5 Expected outputs and evaluation criteria

Expected output (case)	Evaluation criterion (case)
Traffic-class and link-capacity estimator	Accuracy of classification and link estimation
Dynamic QoS policy engine and enforcement module	Reduction in interactive latency, jitter and packet loss
Dashboard: latency, jitter, loss, throughput, queue depth, fairness	Fairness and absence of starvation
Automated baseline-vs-optimised experiments	Adaptation speed and policy stability
API or UI for temporary service intent	CPU overhead, reproducibility and clarity of results
Architecture diagram; setup guide; scripted demo; known limitations	(cross-cutting)

3. Why this matters to Tata Elxsi

- **Fastest-growing vertical.** Media & Communications is about a third of Tata Elxsi's revenue and grew about 22% year on year in Q1 FY27 (company release, 14 July 2026).
- **RDK-B home gateways.** Tata Elxsi builds and ports RDK-B broadband software. Its 2024 RDK-B integration on Qualcomm's 5G fixed-wireless platform lists QoS among the features operators can offer. VeriQ is designed to package as an RDK-B or OpenWrt component.
- **Intent-based, closed-loop operations.** Tata Elxsi's NEURON platform is described as intent-based and closed-loop for operator networks. VeriQ applies the same philosophy at the home edge: intents in, verified outcomes out. This is a thematic fit; integration is not claimed.
- **Services model.** Tata Elxsi grows by attaching accelerators to engineering services. VeriQ is an accelerator that operators could deploy as software on Linux-based CPEs that meet its operating envelope (not every existing gateway).

4. Our solution: VeriQ

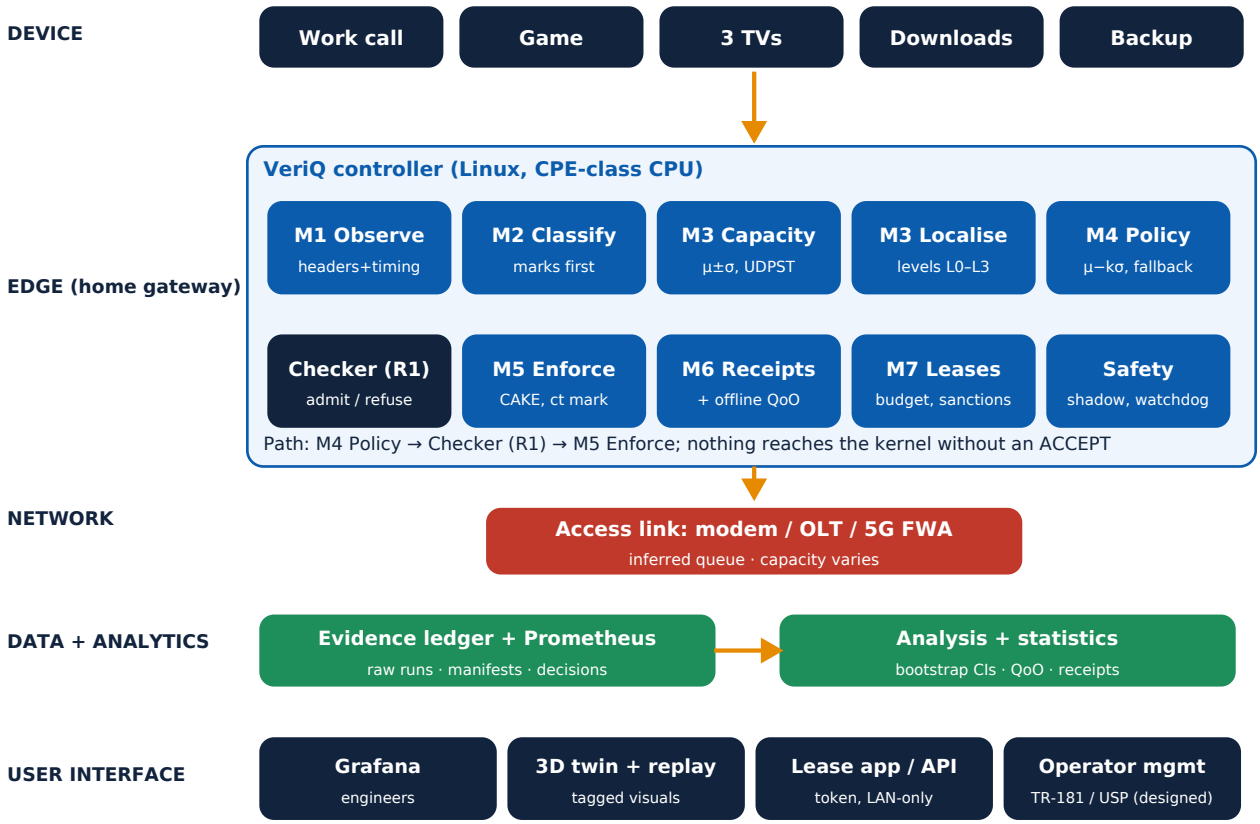
4.1 Design principles

1. **Locate before acting.** No speed cut without evidence that the WAN is the problem; bounded, reversible steps when evidence is unclear.
2. **Deterministic safety, statistical explanation.** Fast guardrails keep delay bounded; slower receipts explain outcomes; only repeated lab experiments support causal claims.
3. **Privacy by construction.** Headers and timing only; app marks first. Classifier abstention alone never grants preference. A valid authenticated lease may grant preference to UNKNOWN traffic, subject to eligibility and operating-state checks.
4. **Honest semantics.** Leases select preferential CAKE treatment. They are not caps or guarantees. Fairness is measured.
5. **Safe rollout.** Shadow mode first, one log row per decision, watchdog to a SAFE shaper.
6. **Every claim has a method.** Each visual is tagged measured, estimated or illustrative; every slide number is recomputed from raw data.

4.2 What VeriQ adds beyond existing tools

Typical approach	VeriQ
Static port/DSCP rules (still common in stock routers)	App marks (NQB, L4S) first, if the flow behaves; else a calibrated classifier that abstains
Delay-triggered rate cuts (cake-autorate, QoSmate autorate; both mature, with reflector handling)	Evidence-graded diagnosis (levels L0–L3) with an action-permission table; designed to hold on remote/Wi-Fi problems (hypothesis, to be tested)
Unchecked rate changes	Deterministic action checker + desired/observed-state reconciler
Fixed shaper	Capacity $\mu \pm \sigma$ tracking; shaper = $\mu - k\sigma$; standard UDPST test on idle lines
Strict priority	Expiring, budgeted leases selecting CAKE tins via conntrack marks (both directions); eligibility violations sanctioned; progress measured
Ping graphs	Real WebRTC freezes and DASH stalls; offline IETF QoO scores with explicit coverage
Set and forget	Receipts: improved / degraded / inconclusive, with condition-aware undo

5. System architecture



The diagram follows the case's required layers: device, edge, network, data, analytics and user interface. The packet path stays in the Linux kernel (tc/CAKE). The controller runs in user space at 1–10 Hz.

5.1 Modules: contracts

Module	Inputs → outputs	Failure handling	Success condition
M1 Observer	Packets (v4/v6) → per-flow size/timing/direction records, DSCP/ECN marks; never payload	Buffer overflow → oldest flows evicted, counted	Zero drops attributable to observer; measured CPU
M2 Classifier	Flow features → class + conformal set; marks first	Uncertain → UNKNOWN (abstention alone never grants preference; a valid lease may)	Coverage, selective accuracy, abstention, macro-F1 under shift
M3 Capacity + localiser	qdisc stats, near/far probes, throughput, UDPST → μ ± σ + diagnosis + confidence	Bad reflector down-weighted; contradiction → UNCERTAIN	E1–E8 estimator tests; diagnosis confusion matrix
M4 Policy / guardrails	Estimate, diagnosis, leases → proposed action	Invariant violation → SAFE; uncertainty → bounded fallback	Settle time, no oscillation, steady-state utilisation
Action Checker (R1)	Proposed action + timestamped snapshot + versions + budgets + leases + state → ACCEPT or REJECT(code)	Refuses: stale data, version mismatch, step above level, below floor, budget exhausted, invalid lease, abstention-only preference, wrong state, duplicate	One unit/fault test per refusal code passes

Module	Inputs → outputs	Failure handling	Success condition
M5 Enforcer	ACCEPTed action → CAKE config + ct-mark rules (ordered, versioned, read back)	Mismatch → compensate; repeated failure → DEGRADED	Applied state equals intended state
M6 Receipts	Before/after windows → improved / degraded / inconclusive (+ offline QoO)	Regime change → INCONCLUSIVE	Receipt matches lab ground truth
M7 Leases	Authenticated requests → grant/refuse with reason	Sanctions override leases and corrections	Leased device improves; others keep progressing
Safety supervisor	Heartbeats, errors → state (SHADOW/LIVE/DEGRADED/SAFE/DISABLED)	Heartbeat loss → watchdog SAFE shaper	Time-to-safe measured (planned, S8)
Evidence harness	Scenario YAML → runs, ledger, reports	Run failure recorded in discrepancy log	Claims check recomputes every published number

6. Key mechanisms

6.1 Fault localisation with corroboration

Near-hop and far-reflector probes are overlapping observations: a far round trip already contains the access link. VeriQ therefore compares **patterns of change** against per-probe baselines rather than adding segments.

Observation	Interpretation	Action
Own queue grows, external delay flat	Gateway is the bottleneck (intended)	None; CAKE manages it
Own queue empty; near and far delay rise	Evidence consistent with a WAN capacity drop	L2 (two independent sources) → fast cut ≤ ×0.75; L0–L1 → bounded fallback only
Only far delay rises	Evidence consistent with a remote-path problem	Hold the shaper
Client-side delay / Wi-Fi retries rise, WAN normal	Wi-Fi problem	Leave WAN shaper; warn
One reflector disagrees	Bad measurement	Down-weight and rotate it out
Contradictory evidence	Uncertain	Hold or bounded fallback; alert on repetition

Evidence levels: L0 no usable signal; L1 one source; L2 two independent sources (all reflectors behind the same access link count as one; e.g. reflector median + throughput censoring with an empty own queue); L3 = L2 + an agreeing capacity test (UDPST, cooperating server). If the near probe is missing, REMOTE cannot exceed L1, and WAN reaches L2 only via the far median + throughput censoring. Estimator accuracy tests are named E1–E8 (a different thing). Full rules: docs/09 Appendix A.7.

6.2 Capacity tracking

- Fast detector initialised from cake-autorate's published defaults (e.g. 3 of 6 samples over a delay threshold; ×0.75 maximum cut; ×1.04 maximum raise; refractory periods), then tuned in the lab.
- Slow estimate $\mu \pm \sigma$ with change-point detection; shaper = $\mu - k\sigma$, so uncertainty buys margin.
- Idle lines: the UDP Speed Test protocol (RFC 9946, April 2026; Broadband Forum TR-471, modelled in TR-181 as IPLayerCapacityMetrics). It needs a cooperating server; without one, tracking is passive and labelled.

6.3 Classification: marks first, honest testing

- DSCP 45 (NQB, RFC 9956) and ECT(1) (L4S) are trusted as hints while the flow behaves.
- Unmarked flows: small calibrated gradient-boosted model after 5/10/20/30 packets; split-conformal sets; act only on single-class sets (a rule for acting, not a claim of correctness).
- Honesty kit, after SIGCOMM 2025 ('The Sweet Danger of Sugar'): flow- and time-based splits, held-out apps, no IP/port/TTL/sequence features, macro-F1, occlusion tests treated as leads, qosify-style and port-rule baselines.

6.4 Enforcement, leases and sanctions

- Core enforcement is CAKE (diffserv4, per-host fairness, overhead compensation).
- **Leases select preferential CAKE treatment. Tin thresholds reduce that preference above configured usage levels. Per-device fairness and other traffic's progress are evaluated separately.** Thresholds are weight changes shared by a tin, not per-device caps

or reservations.

- Admission: at most 2 active leases, one per device, time-limited: a **provisional** rule, not proof of fairness.
- Mark plumbing: the controller writes a private conntrack mark; nftables (upload) and act_ctinfo + IFB (download) turn it into the packet mark, and CAKE's **fwmark** option selects the tin. DSCP is not rewritten by default; per-tin counters and packet captures verify the treatment.
- Precedence: **operating state > sanction > lease > user correction > mark > classifier > default**. A lease covers its device's flows, including UNKNOWN ones (explicit user authorization differs from classifier abstention).
- Sanctions apply only to **violations of preferential-tin eligibility** (a per-flow rate/queue envelope). Ordinary bulk traffic in best effort is legitimate and never sanctioned. Recovery: a penalty period (60 s, doubling, capped at 15 min) with **no observed violation**. Silence counts, so idle apps never have to send traffic to recover.
- Hard guaranteed rates (HTB) and scheduler switching are post-core experiments.

6.5 Receipts and Quality of Outcome

- Compare equal windows only within a stable regime; otherwise INCONCLUSIVE. Block bootstrap; minimum samples.
- Undo is **condition-aware**: re-derive the rate for current capacity rather than restore a stale value.
- Offline QoO (IETF draft-ietf-ippm-qoo-11): same statistic on both sides; sample floors (a few hundred samples are insufficient to characterise p99.9 reliably); coverage label (router probes exclude Wi-Fi); throughput judged only with demand evidence, otherwise **unknown**, and the score is withheld or shown as partial.

6.6 Uncertainty fallback (a hypothesis under test)

When delay rises but the cause is unclear, VeriQ takes small reversible steps. Starting parameters: ≤ 5% per step, ≤ 10% cumulative per rolling 10 minutes (never refunded), 10 s before/after windows, a 60 s cooldown between episodes (it delays the next episode but does **not** reset the cumulative budget), and a throughput guard. A step is kept only if near-probe delay improves with no concurrent change. A queue moving into the router is recorded as **supporting** evidence, not proof. NOT IMPROVED or INCONCLUSIVE steps are undone after re-checking current conditions. Scenario S13 evaluates whether these parameters are safe.

6.7 Safety supervisor

- States: SHADOW → LIVE (human approval) → DEGRADED → SAFE (watchdog fixed shaper); DISABLED cleans up.
- One decision-log row per decision, including 'do nothing', with reason and blocker.
- Failed applies retry with capped exponential backoff. Shadow runs produce disagreement reports, never counts of 'avoided mistakes'.

7. Implementation flow

7.1 Runtime: one control cycle



Fast guardrails act every 0.1–1 s · receipts every 15–60 s · causal claims only from repeated lab experiments

1. Read qdisc statistics, probe deltas, throughput and new flow records.
2. Update classes (marks first, then classifier; apply sanctions).
3. Update capacity μ ± σ; run the fast detector.
4. Localise: own queue / WAN / remote / Wi-Fi / bad probe, with confidence and corroboration.
5. Propose within guardrails: hold, fast cut, slow raise, bounded fallback step, mark or lease change.
6. **Check:** the action checker admits or refuses the proposal; refusals are logged with a code.
7. Apply only ACCEPTed actions (in shadow mode, only record) via ordered, versioned, read-back updates.
8. Open or close a receipt window; record outcome and evidence.
9. Write the decision log row; export metrics to the ledger, Grafana and the 3D replay.

7.2 Build plan (harness first)

Milestone	Scope	Exit criterion
M-A · 25 Oct – 13 Nov	netns lab, scenario runner, **minimal WebRTC call + DASH player + bulk traffic**, evidence ledger, baselines (SQM-CAKE, cake-autorate)	First reproducible call-plus-bulk experiment, 10 runs
M-B · 14–20 Nov	Observer (marks), classifier + honesty kit, capacity estimator, fault localiser, **action checker + refusal tests**	E1–E8 estimator table; diagnosis confusion matrix (S2, S6, S7, S10)
M-C · 21–27 Nov	Guardrails, CAKE enforcer, leases + sanctions, shadow mode, receipts, offline QoO	S1–S5 run end to end in shadow then live
M-D · 28 Nov – 4 Dec	Remaining scenarios, Grafana, 3D replay v1, physical kit bring-up	Tier-2 matrix complete
M-E · 5–13 Dec	Ablations, report, claims check, video, deck	Every published number reproducible from the ledger

7.3 Technology stack

Layer	Choice
Datapath	Linux tc: CAKE (core); netem/tbf for emulation; HTB, DualPI2, eBPF as later experiments
Control plane	Python 3.12 (asyncio) + pyroute2 netlink; systemd units and watchdog
ML	scikit-learn / LightGBM; split-conformal prediction
Measurement	Playwright + Chromium (WebRTC getStats, dash.js), ffmpeg test media, iperf3, irtt, Flent, UDPST
Data / UI	Evidence ledger (raw files + manifests), Prometheus, Grafana, FastAPI, React + three.js replay
Quality	Unit tests per module, scenario CI on Ubuntu runners, claims check, traceability matrix

7.4 Team roles

Role	Owns
Systems / datapath	Lab, harness plumbing, observer, enforcer, safety supervisor, CPU measurement, physical kit
ML / control	Classifier + honesty kit, capacity estimator, localiser, guardrails, statistics, QoO
Platform / experience	Leases API, ledger and reports, Grafana, 3D replay, video, slides

8. Results to date and evaluation plan

8.1 Preliminary result to date (one emulated sandbox run, not yet reproduced)

Setup: four Linux network namespaces (client, router, emulated ISP, server); 20 Mbps down / 5 Mbps up; 300 ms ISP FIFO; four TCP flows for 15 s; ping every 0.1 s under load; added delay = p95 RTT under load − idle median RTT. Sandbox kernel without CAKE, fq_codel, netem or IPv6, so **no base RTT was applied** (an earlier fingerprint wrongly said 20 ms; corrected). Script: lab/quickstart_bufferbloat.sh; raw files and fingerprint: evidence/day0_sandbox_2026-10-11/.

Router behaviour	Download: added delay p95	Upload: added delay p95	Goodput down / up
Stock FIFO	63.6 ms	266.1 ms	19.1 / 4.7 Mbps
Short-buffer shaper (95%)	55.4 ms	39.6 ms	18.1 / 4.4 Mbps
fq_codel, CAKE, VeriQ	Next run on a full kernel		

Reading: the uplink is the worst offender, and simply moving the queue into the router helps dramatically. A blunt short buffer barely helps the download direction, which motivates AQM (CAKE) and VeriQ's adaptive control. This is **one unreplicated run**. It counts as reproduced only after ≥ 10 runs per arm on a full kernel with a regenerated fingerprint.

8.2 Comparison arms and ablations

Arm	What it is	Why
FIFO	Stock router, bloated modem	The problem
fq_codel (no shaper)	Smart queue, wrong place	Shows why shaping matters
SQM-CAKE (static, tuned)	OpenWrt standard	The common fix
cake-autorate (tuned)	Mature open adaptive shaper	Strong rival
QoSmate + autorate (commit pinned)	Mature OpenWrt QoS (CAKE/HFSC/HTB, heuristics, UI)	Strongest whole-system rival
qosify-style rules + CAKE	Heuristic classification	Strongest rule-based classifier
VeriQ	Full system	

Ablations: without the classifier, without the localiser, without leases, without the fallback. These show what each part really adds, including where VeriQ does not beat tuned CAKE.

8.3 Scenarios and targets (targets, not results)

#	Scenario	Target
S1	ISO download during a WebRTC call	0 freezes; added delay p95 ≤ 10 ms; bulk keeps ≥ 85% of leftover
S2	WAN 100 → 20 Mbps mid-call	Correct, corroborated diagnosis; settle < 2 s; no oscillation
S3	3 TVs + gaming	Game added p99 ≤ 20 ms; 0 rebuffers; host Jain index ≥ 0.9
S4	Backup upload during a call	As S1, upstream
S5	Lease 'protect laptop 45 min'	Leased device improves; others keep progressing; expiry on time
S6 / S7	Remote-path delay / lying reflector	Shaper held / reflector ignored
S8	Controller crash	SAFE engaged; time-to-safe measured (planned)
S9	IPv6-only client	Same behaviour as IPv4

#	Scenario	Target
S10 / S11	Wi-Fi bottleneck / false DSCP 45 mark	WAN untouched / flow sanctioned within 1 s
S12 / S13	Shadow beside cake-autorate / ambiguous evidence	Disagreement report / fallback hypothesis evaluated

8.4 Mapping to the case's evaluation criteria

Case criterion	VeriQ evidence
Classification and link-estimation accuracy	Coverage, selective accuracy, abstention, macro-F1 under shift; E1–E8 relative error, range-contains-truth, settle time
Reduction in interactive latency, jitter, loss	Added delay p50/p95/p99, jitter, loss; WebRTC freezes, concealment; DASH stalls; offline QoO
Fairness and absence of starvation	Per-class/host bytes per window; Jain index; sanction behaviour (S11)
Adaptation speed and policy stability	Settle time from injected events; changes per minute; oscillation; diagnosis latency
CPU overhead, reproducibility, clarity	CPU per core on Pi 5 / CPE; evidence ledger + manifests; claims check; receipts in plain English

8.5 Primary research question

Does diagnosis-informed, bounded adaptation reduce inappropriate interventions while staying non-inferior on genuine drops and bulk progress? The primary outcome is measured only in **control-futile** scenarios: remote-path delay, a lying reflector, a Wi-Fi bottleneck, and a probe-path disturbance. These are injected outside what the shaper can usefully control, and the evaluator confirms no local congestion (access utilisation < 80%, ISP queue < 5 ms). Per run we report the **count** of reductions of any size, their **cumulative magnitude** $\int \max(0, S_{ref} - S(t))\,dt$ in Mbit·s, and the **duration** below $0.95 \cdot S_{ref}$. Constant-capacity congestion (e.g. a bulk upload starting) is excluded, because cutting there is correct. Non-inferiority vs tuned cake-autorate: settling confirmation time ≤ +1 s in genuine drops; bulk share of leftover bandwidth ≥ comparator − 5 percentage points (absolute). Faults are scheduled blind: the controller never sees scenario names, true capacity or fault times. Two comparison types: controller-only (same CAKE datapath) and whole-system (each tool as recommended). Two enforcers never run at once.

8.6 Statistics

The experimental unit is a run, not a packet. A pilot (n = 5 per arm) sets the run count; order is counterbalanced with warm-up and reset; run-level medians with bootstrap 95% confidence intervals; Holm correction across comparisons. A tiered matrix starts small (4 arms × 3 scenarios × 10 runs = 120 runs). Every report carries an environment fingerprint and is generated only from the evidence ledger.

9. Business impact and scalability

9.1 Value by stakeholder

Stakeholder	Value
Households	Smoother calls, games and streaming during heavy use, on existing equipment; transparent receipts
Operators / ISPs	Fewer 'slow internet' contacts that speed tests cannot explain; QoE telemetry; a differentiated 'work-from-home ready' service; standards-aligned diagnostics (TR-471 / RFC 9946, TR-181 / USP)
Tata Elxsi	A software accelerator for RDK-B / OpenWrt gateway engineering and managed services; fits the intent-based closed-loop story of NEURON; reusable lab and evidence tooling for customer validations

9.2 Illustrative operator model (assumptions, to be replaced with operator data)

Assumption (illustrative)	Value
Subscribers on VeriQ-enabled gateways (N)	1,000,000
Monthly share contacting support about lag/'slow internet' (r)	1.5%
Cost per support contact (c)	₹400
Share of those contacts VeriQ prevents (p)	15% (to be measured in a field trial)
Gross modelled avoided cost = N × r × 12 × c × p	**≈ ₹1.08 crore per year** (gross)
Net value	Gross − integration − per-device validation/maintenance − support training (all unknown until a pilot)

> **How to present this**
>
> Show the formula and let judges change the inputs. None of these numbers are measured; they illustrate how VeriQ's value would be quantified in an operator pilot.

9.3 Scaling path (designed, not built)

- **Per home:** software-only on CPE-class ARM CPUs; datapath in the kernel; control loop 1–10 Hz; CPU measured in the lab (planned on Raspberry Pi 5 and an ARM OpenWrt router).
- **Packaging:** RDK-B component or OpenWrt/prplOS package; systemd/procd service; install/verify/reset tooling.
- **Fleet:** TR-181 data model + TR-369 USP management; staged rollout that starts every home in shadow mode; aggregated receipts.
- **Later research tracks:** HTB guaranteed rates, NQB lane (RFC 9956), DualPI2/L4S, eBPF datapath, Wi-Fi airtime control, multi-gigabit scaling.

10. Safety, privacy, guarantees and known limitations

10.1 Properties designed and to be tested

1. Never inspects the payload of forwarded traffic. Classifier abstention alone never grants preference. A valid authenticated lease may grant preference to UNKNOWN traffic, subject to eligibility and operating-state checks. Sanctioned flows receive no preference.
2. Never rewrites the DSCP it forwards and sets no ECN bits itself. CAKE may mark ECN-capable packets CE as its normal congestion signal.
3. Leases always expire; sanctions override leases and corrections.
4. Every applied configuration is versioned and read back; every decision is logged.
5. New installs start in shadow mode; a separate watchdog engages a SAFE shaper on controller failure. SAFE assumes capacity does not keep collapsing, so it is not universally safe.
6. The action checker **checks admissibility against the implemented rules** (fresh data, valid version, rate floor and step limits per evidence level, budgets, lease validity, operating state). It does not establish optimality or user benefit.
7. QoO is never reported as complete when a component is unknown.

10.2 Known limitations (stated before judges ask)

- Shaping below capacity costs a few percent of throughput.
- The ISP's queue is inferred, never observed.
- Download control relies on senders responding; unresponsive UDP is only partly controllable.
- Classification without content is probabilistic; many apps do not mark yet.
- Lease preference is not a guarantee; fairness is shown only by measurement.
- The uncertainty-fallback limits are starting hypotheses, not validated safety limits.
- QoO is an IETF draft and our requirement profiles are our own calibration.
- All lab links are emulated; physical-kit and real-line validation are planned.
- CAKE runs per-qdisc on one CPU in the kernels we target. Multi-queue CAKE support depends on the kernel version and is not evidence of VeriQ scaling.
- QoSmate is more mature as a product (UI, packaging, schedulers). VeriQ's advantages are hypotheses until the comparison runs.

11. Demonstration plan and roadmap

11.1 Demonstration (planned)

- **Physical kit:** Raspberry Pi 5 running VeriQ, an emulated-ISP box with a red 'WAN DROP' button, and a Jitsi server **on the WAN side** with peer-to-peer disabled, so call media crosses the impaired link (checked with interface counters). A judge chooses the fault: access drop, misleading remote delay, a stale action (rejected by the checker), or a controller kill.
- **Split-screen video:** the same download on a stock router and on VeriQ, with live freeze counters.
- **3D replay:** a home model whose queues are drawn from recorded telemetry. Every element is tagged measured, estimated or illustrative, and the inferred ISP queue is shown beside lab ground truth.
- **Receipts on screen:** what VeriQ changed, why, and whether calls actually improved.

11.2 Roadmap after the core

Track	Purpose
HTB guaranteed rates + scheduler switching	Real minimum rates for leases (a proposed, unvalidated contract)
NQB lane (RFC 9956)	Shallow, non-priority lane for smooth traffic, with traffic protection
DualPI2 / L4S arm	Compare with the industry's low-latency direction
Header-only call-quality estimator (IMC 2023 method)	Experience estimates in real homes without app access
Live QoO in receipts; lease chat box	After offline calibration; the chat box only fills the lease form

12. References

- Tata Elxsi TELIPORT S4 Case Study 3 problem statement; Tata Elxsi Q1 FY27 press release (14 Jul 2026)
- Tata Elxsi RDK-B on Qualcomm 5G FWA (2024); Tata Elxsi NEURON platform page
- tc-cake manual; cake-autorate (lynxthecat); OpenWrt SQM and qosify
- RFC 9956 NQB PHB (May 2026); RFC 9946 UDP Speed Test Protocol (Apr 2026); Broadband Forum TR-471; TR-181
- draft-ietf-ippm-qoo-11 Quality of Outcome (May 2026); draft-ietf-ippm-responsiveness-09 (Jul 2026)
- Zhao et al., 'The Sweet Danger of Sugar', SIGCOMM 2025; Sharma et al., IMC 2023 (WebRTC QoE without headers)
- Jain & Dovrolis, Pathload / SLoPS (2002); Adams & MacKay, Bayesian Online Changepoint Detection (2007)
- **Authoritative specification: docs/09_VeriQ_v4_Specification.md (v4.1) with Appendix A implementation contracts (rate floor, capacity vs throughput vs goodput, upward exploration, config ownership, first-packet marks, settling onset/confirmation, selection rate vs coverage)**; change record: docs/10; supporting history: docs/05–08